

Science for Students

Grade 8 | AccessLearn Demo Material

Chapter 1: The Solar System

Our solar system is a vast collection of celestial bodies held together by the gravitational pull of the Sun. The Sun, a giant star, sits at the center of our solar system and accounts for more than 99 percent of its total mass. Life on Earth depends on the energy radiated by the Sun in the form of light and heat.

The eight planets of our solar system, in order from the Sun, are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. These planets orbit the Sun in elliptical paths called orbits. The time taken by a planet to complete one orbit around the Sun is called its orbital period or year. Earth takes approximately 365 days to complete one orbit, which defines our calendar year.

The inner planets — Mercury, Venus, Earth, and Mars — are called terrestrial planets because they have solid, rocky surfaces. The outer planets — Jupiter, Saturn, Uranus, and Neptune — are called gas giants or ice giants. They are composed primarily of hydrogen, helium, and other gases and are much larger than the terrestrial planets.

Between Mars and Jupiter lies the asteroid belt, a region filled with millions of rocky objects called asteroids. Beyond Neptune lies the Kuiper Belt, a region rich in icy bodies, including Pluto, which was reclassified as a dwarf planet in 2006. Comets are icy bodies that travel in elongated orbits and develop bright tails of gas and dust when they pass close to the Sun.

Chapter 2: The Water Cycle

The water cycle, also known as the hydrological cycle, describes the continuous movement of water on, above, and below the surface of the

Earth. Water is essential for all forms of life and exists in three states: liquid water in rivers, lakes, and oceans; solid ice in glaciers and snow; and water vapor in the atmosphere.

The water cycle begins with evaporation. The Sun's heat causes water from oceans, rivers, and lakes to evaporate into the atmosphere as water vapor. Plants also release water vapor through a process called transpiration. Together, these processes are called evapotranspiration.

As water vapor rises into the atmosphere, it cools and condenses into tiny water droplets, forming clouds. This process is called condensation. When water droplets in clouds combine and grow large enough, they fall back to Earth as precipitation — in the form of rain, snow, sleet, or hail depending on temperature.

Precipitation that falls on land may flow over the surface as runoff, entering rivers and streams that eventually lead to the ocean. Some water seeps into the ground through a process called infiltration, replenishing underground water stores called aquifers. This groundwater can be accessed through wells and is crucial for drinking and agriculture in many regions.

Chapter 3: Photosynthesis and Respiration

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy from the Sun into chemical energy stored as glucose. This process takes place primarily in the leaves of plants, inside special structures called chloroplasts, which contain a green pigment called chlorophyll.

The chemical equation for photosynthesis is: six molecules of carbon dioxide plus six molecules of water, in the presence of sunlight, produce one molecule of glucose and six molecules of oxygen. This means that plants absorb carbon dioxide from the air and release oxygen as a byproduct — making photosynthesis essential for all oxygen-breathing life on Earth.

Cellular respiration is the opposite of photosynthesis. In respiration, living organisms break down glucose in the presence of oxygen to release energy, carbon dioxide, and water. This energy is stored in a molecule called ATP, or adenosine triphosphate, which powers all cellular activities.

There are two types of respiration: aerobic and anaerobic. Aerobic respiration requires oxygen and produces a large amount of energy. Anaerobic respiration occurs without oxygen, producing less energy, and is used by organisms such as yeast to ferment sugars into alcohol and carbon dioxide. This process is used in baking bread and brewing beverages.

Chapter 4: Force and Motion

A force is a push or pull that acts on an object and can change its speed, direction, or shape. Forces are measured in Newtons, named after the famous scientist Sir Isaac Newton. The net force on an object determines whether the object accelerates, decelerates, or remains in a state of rest or uniform motion.

Newton's First Law of Motion states that an object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction, unless acted upon by an unbalanced external force. This property of objects to resist changes in their state of motion is called inertia. Heavier objects have more inertia than lighter ones.

Newton's Second Law of Motion states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. This is expressed by the formula: Force equals mass multiplied by acceleration. So, for the same force, a lighter object will accelerate more than a heavier object.

Newton's Third Law of Motion states that for every action, there is an equal and opposite reaction. When a swimmer pushes water backward with their hands, the water pushes the swimmer forward. When a rocket expels hot gas downward, the gas pushes the rocket upward into space. These examples show Newton's Third Law in everyday life.